Depropanizer Column Ver 1 · OTS

DC3-V1-PE-HMB-001

Heat and material balance

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
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Title	Heat and material balance
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Originator / DRAWN	Apinan Soottitantawat
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Revision record

REV	DATE	DESCRIPTION	BY	CHK	APP
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Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

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1 · Purpose and scope

This document is the heat and material balance of the C-201 depropanizer unit (feed surge drum V-201, feed pumps P-201A/B, preheater E-201, column C-201 with reboiler E-203, condenser E-202, reflux drum V-202 and pumps P-202A/B). It fixes, for the numbered streams 1–11 of the PFD DC-PFD-200, the flow, composition, temperature, pressure and phase, and the duties of the heat exchangers and pumps, for three cases. It is the basis of the equipment list and datasheets (DC3-V1-ME-EQL-001, -DS-001), the line list (DC3-V1-PE-LIN-001), the relief load basis (DC3-V1-PE-RLB-001), the instrument list and the electrical load list.

Cases

Case	Definition	Feed m³/h	Reflux m³/h	Source
Design case	Design snapshot of the simulator (FIC-201 60 m³/h, reflux 49 m³/h, TIC-208 76.9 °C, PIC-210 1.70 MPa g)	60.0	49.0	snapDesign
Maximum feed	Highest feed inside the operating window with the reflux ratioed to the feed (Design tab: 70.3 m³/h). owRun 14 400 s, trips off.	70.0	57.2	owRun
Turndown	Turndown to 40 m³/h with the reflux ratioed to the feed (window bottom edge 33.5 m³/h). owRun 14 400 s, trips off.	40.0	32.7	owRun

epc/mk_epc.js (design = snapDesign; max/min = owRun 14 400 s, trips off; 7200 s left the drum and inert inventories still moving, mass closure +0.23 / −0.73 %). The pseudo-binary model carries C3 and lighter (light key) and C4 and heavier; mass flows use the assumed molar masses 44.1 / 58.1 kg/kmol and 16.0 kg/kmol for the non-condensables.

2 · Design case

Design snapshot of the simulator (FIC-201 60 m³/h, reflux 49 m³/h, TIC-208 76.9 °C, PIC-210 1.70 MPa g)

Stream	1	2	3	4	5	6	7	8	9	10	11
Description	Feed from the upstream unit	Feed to C-201 (P-201A/B, FV-201)	Preheated feed (partly vaporized)	Overhead vapour	Reflux	Propane product	Vent gas (non-con densables)	Bottoms (C4 and heavier)	LP steam to the feed preheater	LP steam to the reboiler	Cooling water to the condenser
From → to	Battery limit → V-201	V-201 → E-201	E-201 → C-201 tray 18	C-201 top → E-202	P-202A/B → C-201 tray 27	P-202A/B → Storage	V-202 → Fuel gas (PV-210)	C-201 sump → Debutanizer (LV-202)	LP steam header → E-201	LP steam header → E-203	CW supply → E-202 → CW return
Phase	L	L	L+V	V	L	L	V	L	Steam	Steam	Water
Vapour fraction (mol)	0.000	0.000	0.670	1.000	0.000	0.000	1.000	0.000	1.000	1.000	0.000
Temperature °C	−1.0	−1.0	75.0	45.0	39.4	39.4	39.4	109.0	155.8	155.8	15.0 → 25.3
Pressure MPa g	0.400	2.363	1.706	1.700	2.262	2.262	0.400	0.480	0.450	0.450	0.450
Molar flow kmol/h	636.1	636.1	636.1	922.0	555.4	363.4	3.2	269.5	322.9	210.4	18,318.1
Mass flow kg/h	31,809	31,809	31,809	40,798	24,608	16,100	89	15,619	5,817	3,790	330,000
Liquid volume m³/h (meter basis)	60.00	60.00	60.00	—	49.00	32.06	72.5 Nm³/h	27.82	—	—	330.00
C3 and lighter mol % (HC)	57.07	57.07	57.07	98.50	98.50	98.50	98.50	1.00	—	—	—
Non-condensables kmol/h	1.90	1.90	1.90	1.90	—	—	1.90	—	—	—	—
Molar mass kg/kmol	50.01	50.01	50.01	44.25	44.31	44.31	27.66	57.96	18.01	18.02	18.01
Meter (P&ID)	—	FI-201	TI-202	TI-227	FI-204	FI-203	FI-217	FI-202	FI-206	FI-205	FI-216

Duty / key value	Value	Duty / key value	Value
E-201 preheater duty	3,503 kW	Column top pressure / temperature	1.700 MPa g / 45.0 °C
E-203 reboiler duty	2,239 kW	Bottom temperature / tray 8	109.0 / 76.9 °C
E-202 condenser duty	3,956 kW	AI-219 C4+ in propane / AI-220 C3 in bottoms	1.50 / 1.00 mol %
P-201A hydraulic / absorbed	32.3 / 49.7 kW	Flood rectifying / stripping	77 / 74 %
P-202A hydraulic / absorbed	12.4 / 20.7 kW	E-202 blanketed / PV-210 opening	16.9 % / 15 %
Mass closure (1 in; 6 + 7 + 8 out)	31,809 → 31,809 kg/h, 0.001 % ✓	Operating margin (training prices)	5,810 $/h

3 · Maximum feed

Highest feed inside the operating window with the reflux ratioed to the feed (Design tab: 70.3 m³/h). owRun 14 400 s, trips off.

Stream	1	2	3	4	5	6	7	8	9	10	11
Description	Feed from the upstream unit	Feed to C-201 (P-201A/B, FV-201)	Preheated feed (partly vaporized)	Overhead vapour	Reflux	Propane product	Vent gas (non-con densables)	Bottoms (C4 and heavier)	LP steam to the feed preheater	LP steam to the reboiler	Cooling water to the condenser
From → to	Battery limit → V-201	V-201 → E-201	E-201 → C-201 tray 18	C-201 top → E-202	P-202A/B → C-201 tray 27	P-202A/B → Storage	V-202 → Fuel gas (PV-210)	C-201 sump → Debutanizer (LV-202)	LP steam header → E-201	LP steam header → E-203	CW supply → E-202 → CW return
Phase	L	L	L+V	V	L	L	V	L	Steam	Steam	Water
Vapour fraction (mol)	0.000	0.000	0.666	1.000	0.000	0.000	1.000	0.000	1.000	1.000	0.000
Temperature °C	−1.0	−1.0	75.0	45.1	41.8	41.8	41.8	109.1	155.8	155.8	15.0 → 26.8
Pressure MPa g	0.400	2.233	1.708	1.702	2.208	2.208	0.400	0.480	0.450	0.450	0.450
Molar flow kmol/h	742.1	742.1	742.1	1,075.6	647.9	408.3	19.5	314.3	383.1	243.6	18,318.1
Mass flow kg/h	37,111	37,111	37,111	47,608	28,712	18,094	802	18,215	6,902	4,389	330,000
Liquid volume m³/h (meter basis)	70.00	70.00	70.00	—	57.17	36.03	437.2 Nm³/h	32.45	—	—	330.00
C3 and lighter mol % (HC)	57.07	57.07	57.07	98.44	98.44	98.44	98.44	1.05	—	—	—
Non-condensables kmol/h	2.22	2.22	2.22	2.22	—	—	2.22	—	—	—	—
Molar mass kg/kmol	50.01	50.01	50.01	44.26	44.32	44.32	41.10	57.95	18.01	18.01	18.01
Meter (P&ID)	—	FI-201	TI-202	TI-227	FI-204	FI-203	FI-217	FI-202	FI-206	FI-205	FI-216

Duty / key value	Value	Duty / key value	Value
E-201 preheater duty	4,074 kW	Column top pressure / temperature	1.702 MPa g / 45.1 °C
E-203 reboiler duty	2,573 kW	Bottom temperature / tray 8	109.1 / 77.0 °C
E-202 condenser duty	4,547 kW	AI-219 C4+ in propane / AI-220 C3 in bottoms	1.56 / 1.05 mol %
P-201A hydraulic / absorbed	35.2 / 54.1 kW	Flood rectifying / stripping	89 / 85 %
P-202A hydraulic / absorbed	13.0 / 21.7 kW	E-202 blanketed / PV-210 opening	0.0 % / 91 %
Mass closure (1 in; 6 + 7 + 8 out)	37,110 → 37,111 kg/h, 0.002 % ✓	Operating margin (training prices)	6,301 $/h

4 · Turndown

Turndown to 40 m³/h with the reflux ratioed to the feed (window bottom edge 33.5 m³/h). owRun 14 400 s, trips off.

Stream	1	2	3	4	5	6	7	8	9	10	11
Description	Feed from the upstream unit	Feed to C-201 (P-201A/B, FV-201)	Preheated feed (partly vaporized)	Overhead vapour	Reflux	Propane product	Vent gas (non-con densables)	Bottoms (C4 and heavier)	LP steam to the feed preheater	LP steam to the reboiler	Cooling water to the condenser
From → to	Battery limit → V-201	V-201 → E-201	E-201 → C-201 tray 18	C-201 top → E-202	P-202A/B → C-201 tray 27	P-202A/B → Storage	V-202 → Fuel gas (PV-210)	C-201 sump → Debutanizer (LV-202)	LP steam header → E-201	LP steam header → E-203	CW supply → E-202 → CW return
Phase	L	L	L+V	V	L	L	V	L	Steam	Steam	Water
Vapour fraction (mol)	0.000	0.000	0.672	1.000	0.000	0.000	1.000	0.000	1.000	1.000	0.000
Temperature °C	−1.0	−1.0	75.0	44.9	34.8	34.8	34.8	109.0	155.8	155.8	15.0 → 21.9
Pressure MPa g	0.400	2.563	1.704	1.700	2.355	2.355	0.400	0.480	0.450	0.450	0.450
Molar flow kmol/h	424.1	424.1	424.1	614.6	370.3	243.0	1.3	179.7	207.6	143.1	18,318.1
Mass flow kg/h	21,206	21,206	21,206	27,188	16,403	10,765	21	10,421	3,740	2,579	330,000
Liquid volume m³/h (meter basis)	40.00	40.00	40.00	—	32.67	21.44	29.0 Nm³/h	18.56	—	—	330.00
C3 and lighter mol % (HC)	57.07	57.07	57.07	98.62	98.62	98.62	98.62	0.89	—	—	—
Non-condensables kmol/h	1.27	1.27	1.27	1.27	—	—	1.27	—	—	—	—
Molar mass kg/kmol	50.01	50.01	50.01	44.24	44.29	44.29	16.56	57.97	18.02	18.02	18.01
Meter (P&ID)	—	FI-201	TI-202	TI-227	FI-204	FI-203	FI-217	FI-202	FI-206	FI-205	FI-216

Duty / key value	Value	Duty / key value	Value
E-201 preheater duty	2,340 kW	Column top pressure / temperature	1.700 MPa g / 44.9 °C
E-203 reboiler duty	1,546 kW	Bottom temperature / tray 8	109.0 / 76.9 °C
E-202 condenser duty	2,641 kW	AI-219 C4+ in propane / AI-220 C3 in bottoms	1.39 / 0.90 mol %
P-201A hydraulic / absorbed	23.8 / 36.6 kW	Flood rectifying / stripping	53 / 51 %
P-202A hydraulic / absorbed	9.5 / 15.9 kW	E-202 blanketed / PV-210 opening	49.3 % / 6 %
Mass closure (1 in; 6 + 7 + 8 out)	21,206 → 21,207 kg/h, 0.004 % ✓	Operating margin (training prices)	3,898 $/h

5 · Duties, pumps and how the numbers were obtained

Item	Quantity	Design case	Maximum feed	Turndown
E-201	kW	3,503	4,074	2,340
E-203	kW	2,239	2,573	1,546
E-202	kW	3,956	4,547	2,641
P-201A hydraulic	kW	32.3	35.2	23.8
P-201A absorbed	kW	49.7	54.1	36.6
P-202A hydraulic	kW	12.4	13.0	9.5
P-202A absorbed	kW	20.7	21.7	15.9
P-201 flow	m³/h	60.0	70.0	40.0
P-201 ΔP	kPa	1,940	1,810	2,140
P-201 head	m	360	336	398
FV-201 opening	%	40	60	21
P-202 flow	m³/h	81.1	93.2	54.1
P-202 ΔP	kPa	552	504	634
P-202 head	m	119	110	135

Method

- Liquid flows on the flow meters' basis: m³/h ÷ model molar volume (C3 0.088, C4 0.1034 m³/kmol, mixed by mole fraction) = kmol/h; × molar mass = kg/h.
- Duties are the model's: E-201 and E-203 by the steam-side bisection (chest pressure where the steam valve flow equals UA·ΔT), E-202 = condensed vapour × latent heat 15 500 kJ/kmol; checked against the cooling water (330 m³/h × 4.19 kJ/kg K × ΔT).
- Pumps: hydraulic power = Q × ΔP from the model's pump curves (P-201: $\Delta P = 2300 - 0.1 \cdot Q^2$; P-202: $\Delta P = 700 - 0.0226 \cdot Q^2$); absorbed = hydraulic ÷ η (assumed). Rated point 110 % of the maximum-feed flow: P-201 77.0 m³/h, 56.2 kW → motor 75 kW; P-202 102.5 m³/h, 22.0 kW → motor 30 kW.
- Vent gas: the non-condensables entering with the feed leave with the vent (steady state); the rest of the vent is hydrocarbon at the overhead composition.
- Off-design cases run 14 400 s on copies of the model with the trips disabled; 7 200 s left the reflux-drum and non-condensable inventories still moving (mass closure +0.23 / −0.73 %).

Basis and assumptions

Feed and products

- Feed: 60 m³/h, 57.07 mol % C3 and lighter, −1 °C from V-201 at 0.40 MPa g. Pseudo-binary C3 and lighter (light key) / C4 and heavier, plus 0.3 mol % non-condensables (methane / hydrogen) on top of the hydrocarbon feed.
- Propane product (stream 6): C4 and heavier ≤ 2.5 mol % (AI-219)
- Bottoms to the debutanizer (stream 8): C3 and lighter ≤ 2.0 mol % (AI-220)
- Vent gas (stream 7): to the fuel-gas system at 0.40 MPa g through PV-210

Utilities

- LP steam: header 450 kPa g, saturated 155.8 °C (model); design 0.70 MPa g / 175 °C (assumed)
- Cooling water: supply 15 °C, 330 m³/h to E-202 (model); supply pressure 0.45 MPa g, return ≥ 0.25 MPa g, design 0.70 MPa g / 75 °C (assumed)
- Fuel gas: vent receiver at 0.40 MPa g (model)
- Flare: PSV-201 discharge; back-pressure not modelled (assumed atmospheric for sizing)
- Instrument air: fail-safe positions as on the P&ID; 0.6 MPa g dry air (assumed)
- Electric power: 400 V 3-phase 50 Hz motors ≤ 200 kW; 230 V UPS for DCS / SIS (assumed)

Site

- Ambient design dry bulb: 40 °C maximum, 15 °C minimum (assumed, tropical coastal site)
- Maximum metal temperature of uninsulated LPG vessels in the sun: 50 °C (assumed)
- Relative humidity: 80 % (assumed)
- Wind and seismic: not assessed (assumed low-seismic coastal site; to be confirmed by civil)

Physical properties and calculation basis (assumed)

- Molar masses: C3 and lighter 44.1, C4 and heavier 58.1, non-condensables 16.0 kg/kmol (assumed, as the Design tab; 16.0 for methane)
- Liquid density: model molar volumes 0.088 / 0.1034 m³/kmol (501 / 562 kg/m³) with −1.25 / −1.08 kg/m³ per K above 15 °C (assumed; gives the Design tab's 465 / 460 kg/m³ at the top and bottom)
- Vapour density: real gas, Z 0.73 (top) to 0.68 (bottom) as the Design tab, linear in the light-key fraction (assumed)
- Viscosity: hydrocarbon liquid 0.10 mPa s, hydrocarbon vapour 0.010, steam 0.014, water 1.0 mPa s (assumed)
- Pipe: ASME B36.10 Sch 40 inside diameters, roughness 0.046 mm (assumed)
- m³/h in the HMB: liquid volume on the model's basis (what FI-201, FI-202, FI-203 and FI-204 read); velocities use the actual density at the line temperature
- Pumps: efficiencies 65 % (P-201) and 60 % (P-202), rated flow 110 % of the maximum-feed case, NPSHr 3.0 m with 1.0 m margin (assumed).
- Thermosyphon E-203 outlet 30 % mass vaporized (assumed; circulation not modelled).
- Operating window from the Design tab (tabs/design/work/ow_bounds.js (owRun, 7200 s runs, trips off)): feed with the reflux ratioed 33.5–70.3 m³/h.

Codes and practices

- ASME BPVC VIII Div. 1 — pressure vessels and exchanger shells (UG-27, UG-32, UG-99)
- ASME B31.3 — process piping
- ASME B16.5 — flange pressure–temperature ratings
- TEMA (class R) — shell-and-tube exchangers
- API 610 — centrifugal pumps (motor margin, rated point)
- API 520 / 521 — relief sizing and relief scenarios
- API 2510 — LPG installations (design pressure from vapour pressure at the maximum product temperature)
- IEC 60079-10-1 — hazardous-area classification
- IEC 61511 — SIS (see the SRS of SIF tab)

Engineering queries raised

Findings made while generating this document. They are open until answered by the discipline named; the Project & deliverables tab collects them for the whole unit.

No.	Subject	Query	Proposed answer	Status
EP-01	PSV-201 set pressure vs operating pressure	PSV-201 is set at 1.80 MPa g, only 5.9 % above the 1.70 MPa g operating pressure (≥ 10 % is usual; the maximum-feed case runs at 1.702 MPa g). Design-pressure rule (1) gives 1.90 MPa g for C-201 / V-202.	Set PSV-201 at the design pressure 1.90 MPa g (margin 11.8 %) and re-check the relief load and orifice in the Relief tab; or keep 1.80 MPa g and accept simmering risk with a pilot-operated valve.	OPEN
EP-02	Line 6"-P-205 outside the sizing criteria	6"-P-205 (Condensate to the reflux drum (gravity)) fails the “Liquid, gravity drain (self-venting)” criteria in the design case / maximum feed: v > 1 (v 1.29 m/s, ΔP 3.99 kPa/100 m); v > 1, ΔP > 5 (v 1.49 m/s, ΔP 5.28 kPa/100 m).	Change to 8" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-03	Line 1"-P-206 outside the sizing criteria	1"-P-206 (Non-condensable vent) fails the “Vapour, vent gas” criteria in the maximum feed: ΔP > 25 (v 10.5 m/s, ΔP 180.33 kPa/100 m).	Change to 1.5" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-04	Line 4"-P-208 outside the sizing criteria	4"-P-208 (Reflux / product pump discharge header) fails the “Liquid, pump discharge / pressure flow” criteria in the maximum feed: v > 3 (v 3.37 m/s, ΔP 43.64 kPa/100 m).	Change to 6" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-05	Line 8"-P-211 outside the sizing criteria	8"-P-211 (Thermosyphon reboiler feed) fails the “Liquid, thermosyphon reboiler inlet” criteria in the design case / maximum feed: v > 1.5, ΔP > 5 (v 1.87 m/s, ΔP 5.84 kPa/100 m); v > 1.5, ΔP > 5 (v 2.15 m/s, ΔP 7.68 kPa/100 m).	Change to 10" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-06	Line 6"-LS-301 outside the sizing criteria	6"-LS-301 (LP steam to the feed preheater) fails the “LP steam” criteria in the design case / maximum feed: v > 30 (v 30.18 m/s, ΔP 13.38 kPa/100 m); v > 30 (v 35.81 m/s, ΔP 18.7 kPa/100 m).	Change to 8" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-07	Condenser E-202 capacity at maximum feed	In the maximum-feed case (70 m³/h) the non-condensables are swept out (blanketing 0 %), PV-210 is at 90.8 % and the vent carries 17.3 kmol/h of hydrocarbon to fuel gas (design 1.3). E-202 has no capacity left.	Treat ≈ 70 m³/h as the hydraulic/thermal limit of the unit in the design basis, or add condenser area (≥ 10 % margin) if the feed is to rise.	OPEN
EP-08	Product rundown and condensate lines not numbered	The propane rundown downstream of the reflux tee and LV-203, and the steam-condensate lines after ST-201 / ST-202, have no line numbers on DC3-V1-PID-001 rev D.	Number them 3"-P-213, 2"-CD-501 and 2"-CD-502 (sizes from the line list) in the next P&ID revision.	OPEN
EP-09	Thermosyphon circulation not in the model	The model computes the reboiler boil-up (520 kmol/h) but not the thermosyphon circulation; lines 8"-P-211 and 10"-P-212 were sized with 30 % mass vaporized at the outlet (assumed).	Confirm the circulation ratio with a thermosyphon rating (static head, outlet quality 15–35 %) before freezing the reboiler piping.	OPEN
EP-10	Volume basis of the flow meters	The model's liquid molar volumes are temperature independent (0.088 / 0.1034 m³/kmol); the HMB m³/h are therefore on a 15 °C basis while the hot bottoms (109 °C) occupy ≈ 22 % more volume.	State the reference temperature of every FI range and use actual density for meter sizing (instrument data sheets).	OPEN
EP-11	E-201 steam chest pressure at turndown	At turndown (40 m³/h) TV-202 throttles the E-201 shell to 6.7 kPa g (design 148.1): condensate cannot drain through ST-201 against a condensate header at a few hundred kPa g, the shell floods (stall) and the preheat control becomes erratic.	Size ST-201 for the stall condition: pumping trap or condensate pot with level control, or put TV-202 on the condensate side; state the condensate-header back-pressure in the basis.	OPEN
EP-12	NPSH of P-201 relies on the V-201 pressure	NPSHa of P-201 is 20.2 m with V-201 at 0.40 MPa g (feed bubble pressure 312 kPa g at -1 °C). If the drum liquid were at its bubble point only the static head is left: 3.9 m against NPSHr 3 m + 1 m margin.	Confirm how the V-201 pressure is held above the bubble pressure (upstream pressure / blanketing) or raise V-201 by ≥ 0.2 m; give the pump vendor the bubble-point case.	OPEN